



ACHARYA INSTITUTE OF TECHNOLOGY

Affiliated to Visvesvaraya Technological University, Belagavi, Govt. of Karnataka.

Approved by AICTE, New Delhi

Computer Science & Engineering (Data Science)

PROJECT REPORT ON

From BI to Big Data : Explain, Design & Defend

Subject Name: Big Data Analytics

Subject Code: BAD601

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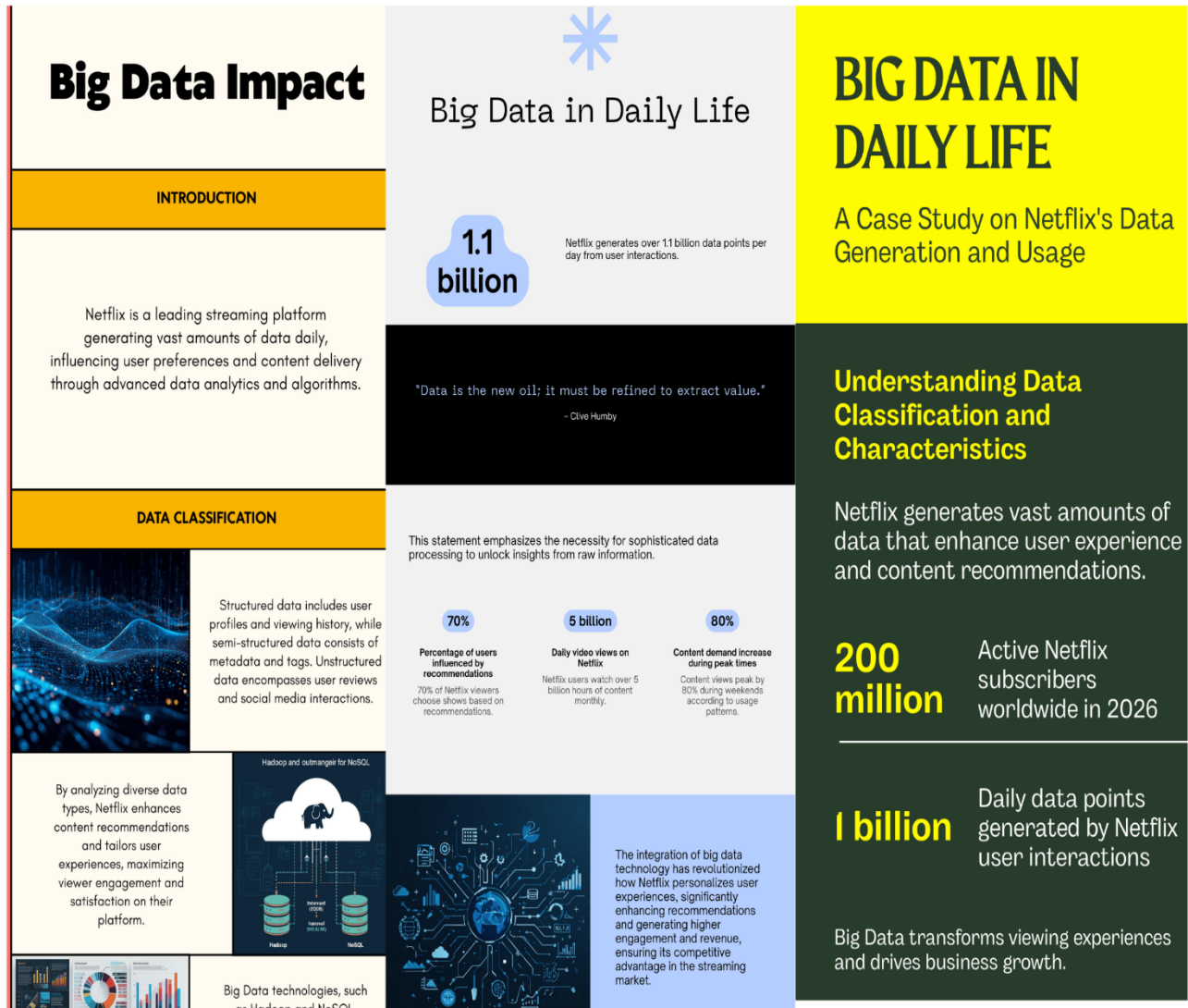
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TASK 1 : Big Data in Daily Life – Visual Storytelling





Big Data in Daily Life

A Case Study on Netflix: Analyzing data generation and its implications for user experience and recommendations.

Types of Data

- Structured Data: User profiles
- Semi-Structured Data: Viewing history
- Unstructured Data: Reviews and ratings

200 million

Netflix subscribers worldwide as of early 2026.

62%

Of users rely on personalized recommendations for viewing choices.

TRANSFORMING VIEWING EXPERIENCES THROUGH DATA

BIG DATA

Impact – Extended Analysis

CHARACTERISTICS

The 5 V's of Big Data are Volume, Velocity, Variety, Veracity, and Value, each defining its unique nature.



LIMITATIONS

Traditional tools like Excel and SQL struggle with large-scale **streaming data**, limiting analysis and insight generation capabilities.

NEED

Big Data technologies like Hadoop and Spark enable **real-time analytics** and handle vast amounts of unstructured data effectively.



EXPLORE SOLUTIONS

Discover how Big Data can transform your business processes.

Big Data Insights

Impact Metrics

Key Performance Indicators

▲ METRIC 1

85%

Data-driven decision-making has improved efficiency significantly.



▲ METRIC 2

70%

Increased customer engagement through tailored solutions and analytics.



▲ METRIC 3

65%

Real-time analytics have enhanced operational



TASK 2 : BI vs Big Data Play

From Traditional BI to Big Data: A Technical Conversation

In this discussion, a Business Manager believes that traditional Business Intelligence tools such as Excel, SQL databases, and Data Warehouses are sufficient.

Conversation

Business Manager: We already use SQL databases and a centralized Data Warehouse for analytics. Why should we move to Big Data technologies?

Data Consultant: Traditional SQL systems are optimized for structured data and OLTP workloads. However, modern businesses generate high-volume, high-velocity, and high-variety data that exceeds the limitations of centralized architectures.

Business Manager: Our Data Warehouse handles OLAP queries efficiently. Isn't that enough?

Data Consultant: Data Warehouses are schema-based and primarily designed for structured relational data. They struggle with semi-structured formats like JSON logs and unstructured data such as videos, images, and clickstream data.

Business Manager: Can't we simply scale our relational database vertically?

Data Consultant: Vertical scaling increases hardware capacity on a single machine, which is costly and limited. Big Data systems adopt horizontal scaling by distributing data across multiple nodes in a cluster.

Business Manager: What exactly does Hadoop contribute?

Data Consultant: Hadoop provides HDFS, a distributed file system that stores massive datasets across multiple machines with fault tolerance. It allows large-scale parallel data processing.

Business Manager: How is data processed in such systems?

Data Consultant: Initially through MapReduce for distributed batch processing, and more efficiently today using Apache Spark, which enables in-memory parallel computation for faster analytics.

Business Manager: Why introduce NoSQL databases?

Data Consultant: NoSQL databases support schema-less designs, allowing flexible storage of semi-structured and unstructured data formats. This flexibility is crucial for modern dynamic datasets.

Business Manager: What about data consistency and reliability?

Data Consultant: Distributed systems follow CAP theorem trade-offs. Big Data platforms balance consistency, availability, and partition tolerance depending on application requirements.

Business Manager: How does Big Data improve decision-making?

Data Consultant: It enables real-time analytics, predictive modelling, machine learning integration, demand forecasting, and personalization at scale.

Business Manager: So Big Data replaces traditional BI?

Data Consultant: Not replaces — it extends BI capabilities. Traditional BI supports descriptive analytics, while Big Data enables scalable, real-time, and advanced analytics.

Business Manager: I understand now. Big Data is necessary for modern data-driven enterprises.

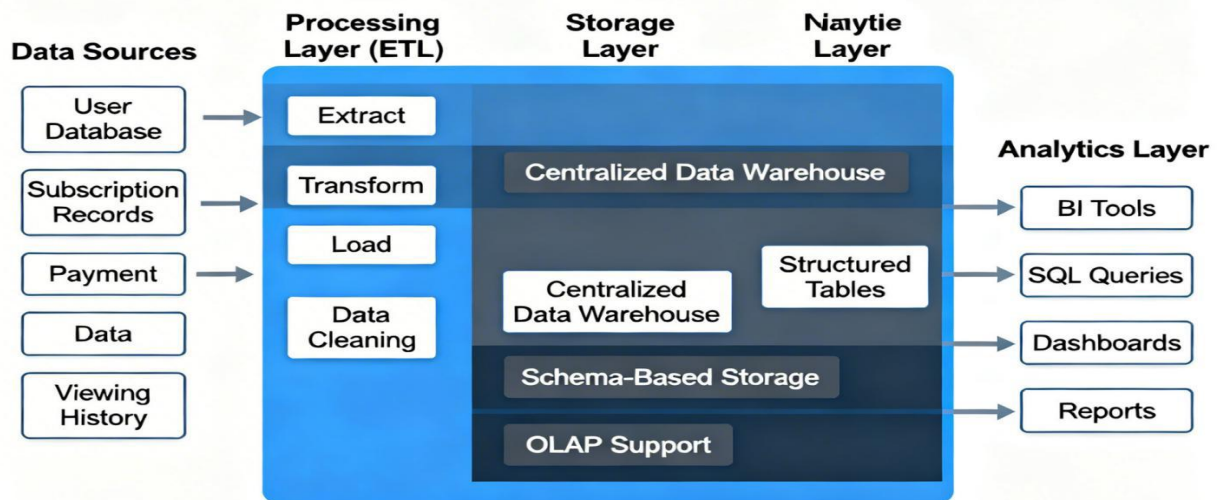
Data Consultant: Exactly. Organizations leveraging distributed architectures gain scalability, flexibility, and competitive advantage.

TASK 3 : Architecture Design Challenge

Traditional Data Warehouse:

A Traditional Data Warehouse architecture is a centralized data management system designed to store and analyse structured data collected from multiple operational sources. It primarily supports Business Intelligence (BI) activities such as reporting, dashboard creation, and historical data analysis. The architecture mainly supports descriptive analytics and batch processing. However, it has limitations in handling unstructured data, real-time streaming data, and large-scale distributed workloads.

Traditional Data Warehouse Architecture - Netflix Case Study

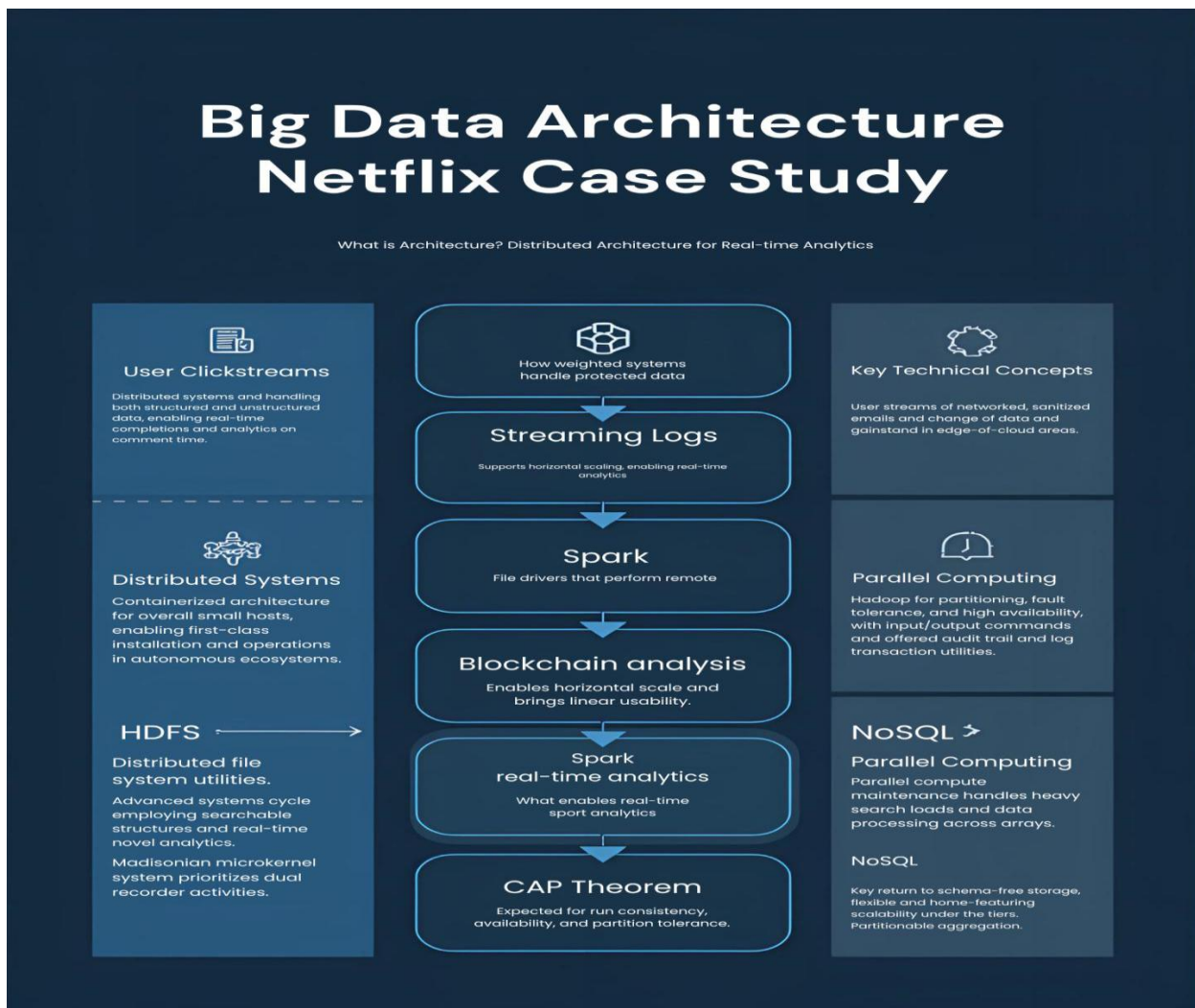


Limitations of Traditional Architecture

- Handles mainly structured data
- Vertical scaling
- Batch processing
- Limited scalability
- Not suitable for real-time analytics

Hadoop-based Big Data Architecture

Hadoop-Based Big Data Architecture is a distributed data processing framework designed to handle massive volumes of structured, semi-structured, and unstructured data. Unlike traditional centralized systems, this architecture distributes storage and computation across multiple nodes to ensure scalability and fault tolerance. This architecture enables horizontal scaling, high performance, and real-time data-driven decision-making.



TASK 4 : Analytics and Tool Match

Descriptive Analytics

Answers: What happened?

- Summarizes historical data using reports and dashboards.
- Identifies trends and patterns from past data.

Diagnostic Analytics

Answers: Why did it happen?

- Analyses data to find the root cause of a problem or trend.
- Uses comparison and correlation analysis to explain changes.

Predictive Analytics

Answers: What will happen next?

- Uses historical data and machine learning to forecast future outcomes.
- Identifies patterns to predict trends and user behaviour.

Prescriptive Analytics

Answers: What action should be taken?

- Recommends the best action based on predictive results.
- Uses optimization and AI models to improve decision-making.

Business Question	Analytics Type	Tool Used
What happened?	Descriptive Analytics	Hadoop + BI Tools
Why did it happen?	Diagnostic Analytics	Spark + Data Mining
What will happen next?	Predictive Analytics	Spark MLlib / Machine Learning
What action should be taken?	Prescriptive Analytics	AI Recommendation Engine + NoSQL

Table 4.1 – Analytics and Tools Table

BONUS CHALLENGE : Explain Big Data to a 10-Year-Old

1. What is Data?

Data is just information.

For example:

- Your name
- Your marks
- Your favourite cartoon
- The videos you watch

All these are called data.

2. What is Big Data?

Imagine:

- One student watching YouTube → Small data
- One school watching YouTube → Bigger data

- The whole world watching YouTube at the same time → HUGE data

That huge amount of information is called Big Data.

Big Data means a very large amount of information that grows very fast.

3. Why is it “Big”?

Big Data is special because:

1. It is Very Large (millions of users)
2. It is Very Fast (created every second)
3. It comes in Different Forms (videos, text, images, clicks)

This is called the 5 Vs of Big Data:

- Volume (Huge size)
- Velocity (High speed)
- Variety (Different types)
- Veracity (Accuracy matters)
- Value (Useful information)

4. Why Not Just Use Excel?

Excel is good for:

- School marks
- Small shop bills

But Big Data is:

- Too big
- Too fast
- Too complex

Normal tools may:

- Become slow
- Hang
- Crash

So, companies need powerful systems.

5. How Big Data Helps Companies

For example, Netflix or YouTube:

Big Data helps them:

- Suggest videos you may like
- Show trending content
- Improve video quality
- Decide what new shows to create

Without Big Data:

- No recommendations
- Random videos
- Poor experience

Big Data is like a giant smart brain that remembers what millions of people do and helps companies make better decisions quickly. Big Data is not just “lots of data.” It is a smart way of handling huge information to make better decisions.